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Advanced Computer Architecture

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Name
Last Name

Pick 2 out of th 3 problems, DO NOT DO ALL THE 3!

Questions (from Forms) (20%)	
Problem 1 (40%)	
Problem 2 (40%)	
Problem 3 (40%)	
Total (100%)	

Problem 1

Assume that the following code is executed on a CPU with SCOREBOARD and with the following units:

- 1 LOAD/STORE unit with Latency= 1
- 2 MULT units with latency = 10
- 1 DIVIDE unit with latency = 30
- 1 ADD/SUBD unit with latency 2

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	5	6	7	8
MULTD F0 F4 F2	6	9	19	20
ADD F8 F2 F6	7	9	10	11
DIVD F12 F8 F0	8	21	51	52
SUBD F8 F6 F2	9	13	14	15

1 LDU 1 CC
2 MULT 10
1 DIV 30
1 ADD/SUB 2

A. List all the possible conflicts in the code.

B. Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?

C. If the previous table was not correct, please, write the right one by having:

- 1 LOAD: 1cc
- 2 MULT: 10cc
- 1 DIVL 30cc
- 3 ADD/SUB: 1cc

Answer 1.A

RAW F6 I1-I4
RAW F6 I1-I6

RAW F2 I2-I3
RAW F2 I2-I4
RAW F2 I2-I6

RAW F0 I3-I5
RAW F8 I4-I5

WAR F8 I5-I6
WAW F8 I4-I5

Answer 1.B

No for several reasons, among which WAW and WAR unsolved, and the given latency of the FU is not respected.
One of those is enough to say that no.

Answer 1.C

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	5	6	7	8
MULTD F0 F4 F2	6	9	19	20
ADD F8 F2 F6	7	9	10	11
DIVD F12 F8 F0	8	21	51	52
SUBD F8 F6 F2	12	13	14	22

Problem 2

Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 0	Exclusive (0)	Invalid (1)	Yes	Yes
1	P2: write block 1	Exclusive (0)	Mod (1)	Yes	No
2	P1: write block 0	Mod (0)	Mod (1)	No	No
3	P2: read block 1	Mod (0)	Mod (1)	No	No
4	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
5	P2 : write block 1	Invalid (1)	Mod (1)	Yes	No
6	P1: read block 0	Ex (0)	Mod (1)	Yes	No
7	P2: write block 1	Ex (0)	Mod (1)	Yes	No
8	P1: read block 0	Ex (0)	Mod (1)	Yes	No
9	P2: read block 1	Ex (0)	Mod (1)	Yes	No

Problem 3

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.
(R0 is set to 10, R1 is set to 0)

```

      LOOP:      LD      F3  0   R0
                  ADDD    F1  F3  F3
                  MULTD   F2  F3  F1
                  ADDI     R1  R1  10
      LOOP2:     LD      F3  0   R1
                  MULTD   F2  F2  F3
                  SUBI     R1  R1  1
                  BNEZ     R1  LOOP2
                  SUBI     R0  R0  10
                  BNE      R0  R1  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

R0 is set to 10, R1 is set to 0

By entering LOOP, R1 is therefore = to 10

This implies that LOOP2 is iterated 10 times

After this first 10 iterations of LOOP2 R0 is = to 0, which means that LOOP is not iterated.

Within this context, we don't care about the possibility of having the same lane pointed by both the loops, because the worst initialization is the same.

1-BHT:

WORST case = LOOP: T LOOP2: NT

1st- LOOP2: miss

8 – LOOP2: correct

10th – LOOP2: miss

LOOP: miss

Miss: 27%

BEST case = LOOP: NT LOOP2: T

9 – LOOP2: correct

10th – LOOP2: miss

LOOP: correct

Miss: 9%

2-BHT:

WORST case = LOOP: ST LOOP2: SNT

1-2 - LOOP2: miss

7 - LOOP2: correct

10th - LOOP2: miss

LOOP: miss

Miss: 36%

BEST case = LOOP: SNT LOOP2: ST

9 - LOOP2: correct

10th - LOOP2: miss

LOOP: correct

Miss: 9%

Therefore the 1-BHT is outperforming, not as expected in "theory", the 2-BHT.
Or, considering that we are not working with "nested loops", the results is inline with the theory, which still means that the 1-BHT is outperforming the 2-BHT.